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SUPPLEMENTARY INFORMATION

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Primate mosaic brain evolution reflects selection on sensory and cognitive specialization

Alex R. DeCasien[1,2*] and James P. Higham[1,2]

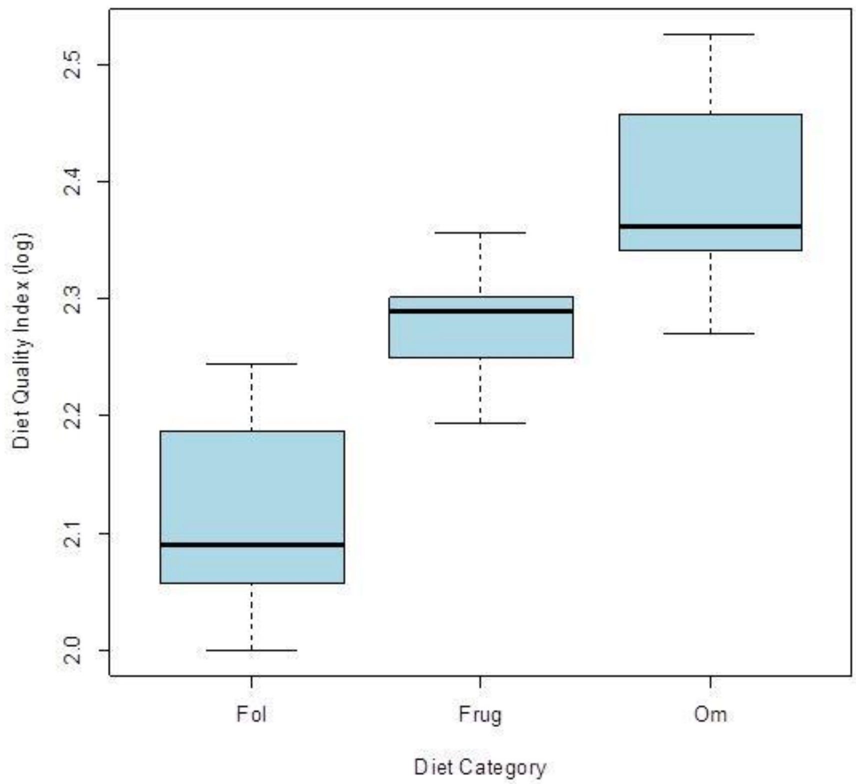
[1]Department of Anthropology, New York University, New York, NY, USA. [2]New York Consortium in Evolutionary Primatology, New York, NY, USA. *e-mail: alex.decasien@nyu.edu

Supplementary Materials for 'Primate mosaic brain evolution reflects selection on sensory and cognitive specialization'

Alex R. DeCasien* and James P. Higham
*Corresponding author. E-mail: alex.decasien@nyu.edu

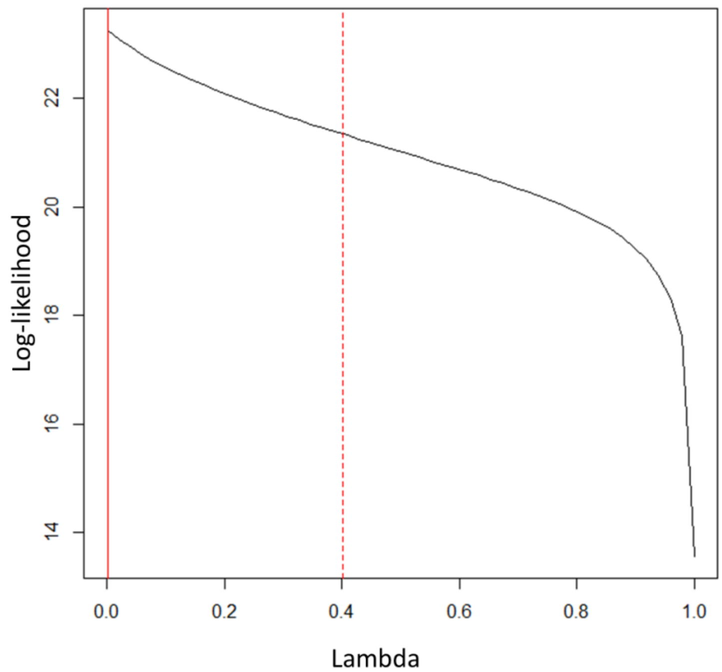
Supplementary Methods

Type III (marginal) ANOVA is an appropriate method for analyzing individual PGLS models when designs are unbalanced and/or predictors are not orthogonal[121]. This test was applied to each equivalent best fit model (dBIC<2) to identify which predictors explain a significant amount of variation in the volume of each region across species. These results are available in the Supplementary Tables.



Supplementary Figure 1 | Differences in DQI between diet categories (spp. n=61): DQI (log) ~ Diet Category

Diet quality index (log) differs significantly between diet categories ($p<0.0001$). Omnivorous species exhibit the highest DQIs, followed by frugivores and then folivores (omnivores > frugivores: $p<0.0001$; omnivores > folivores: $p<0.0001$; frugivores > folivores: $p<0.0001$).



Supplementary Figure 2 | Log-likelihood plot of lambda for the insula model (spp. n=30): Region (log) ~ ROB (log) + Suborder + Diet Category + System (*Pongo* = solitary) + Activity Pattern (*Aotus* = nocturnal)

Supplementary Tables

1. **10k Trees consensus tree BIC comparisons:**

   Tables S1-S12.xlsx

   a. Model = Diet category + System (*Pongo*=solitary) + Activity (*Aotus*=cathemeral):

      ‘Diet + System (P=S) + Act (A=C)’ tab (Table S1)

   b. Model = Diet category + Group size (log) + Activity (*Aotus*= cathemeral):

      ‘Diet + Group Size + Act (A=C)’ tab (Table S2)

   c. Model = DQI (log) + System (*Pongo*=solitary) + Activity (*Aotus*= cathemeral):

      ‘DQI + System (P=S) + Act (A=C)’ tab (Table S3)

   d. Model = DQI (log) + Group size (log) + Activity (*Aotus*= cathemeral):

      ‘DQI + Group Size + Act (A=C)’ tab (Table S4)

   e. Model = Diet category + System (*Pongo*=group) + Activity (*Aotus*= cathemeral):

      ‘Diet + System (P=G) + Act (A=C)’ tab (Table S5)

   f. Model = DQI (log) + System (*Pongo*=group) + Activity (*Aotus*= cathemeral):

      ‘DQI + System (P=G) + Act (A=C)’ tab (Table S6)

   g. Model = Diet category + System (*Pongo*=solitary) + Activity (*Aotus*=nocturnal):

      ‘Diet + System (P=S) + Act (A=N)’ tab (Table S7)

   h. Model = Diet category + Group size (log) + Activity (*Aotus*= nocturnal):

      ‘Diet + Group Size + Act (A=N)’ tab (Table S8)

   i. Model = DQI (log) + System (*Pongo*=solitary) + Activity (*Aotus*= nocturnal):

      ‘DQI + System (P=S) + Act (A=N)’ tab (Table S9)

   j. Model = DQI (log) + Group size (log) + Activity (*Aotus*= nocturnal):

      ‘DQI + Group Size + Act (A=N)’ tab (Table S10)

k. Model = Diet category + System (*Pongo*=group) + Activity (*Aotus*= nocturnal):

   ‘Diet + System (P=G) + Act (A=N)’ tab (Table S11)

l. Model = DQI (log) + System (*Pongo*=group) + Activity (*Aotus*= nocturnal):

   ‘DQI + System (P=G) + Act (A=N)’ tab (Table S12)

2. **10k Trees consensus tree PGLS/ANOVA results:**

   Tables S13-S24.xlsx

   a. Model = Diet category + System (*Pongo*=solitary) + Activity (*Aotus*=cathemeral):

      ‘Diet + System (P=S) + Act (A=C)’ tab (Table S13)

   b. Model = Diet category + Group size (log) + Activity (*Aotus*= cathemeral):

      ‘Diet + Group Size + Act (A=C)’ tab (Table S14)

   c. Model = DQI (log) + System (*Pongo*=solitary) + Activity (*Aotus*= cathemeral):

      ‘DQI + System (P=S) + Act (A=C)’ tab (Table S15)

   d. Model = DQI (log) + Group size (log) + Activity (*Aotus*= cathemeral):

      ‘DQI + Group Size + Act (A=C)’ tab (Table S16)

   e. Model = Diet category + System (*Pongo*=group) + Activity (*Aotus*= cathemeral):

      ‘Diet + System (P=G) + Act (A=C)’ tab (Table S17)

   f. Model = DQI (log) + System (*Pongo*=group) + Activity (*Aotus*= cathemeral):

      ‘DQI + System (P=G) + Act (A=C)’ tab (Table S18)

   g. Model = Diet category + System (*Pongo*=solitary) + Activity (*Aotus*=nocturnal):

      ‘Diet + System (P=S) + Act (A=N)’ tab (Table S19)

   h. Model = Diet category + Group size (log) + Activity (*Aotus*= nocturnal):

      ‘Diet + Group Size + Act (A=N)’ tab (Table S20)

i. Model = DQI (log) + System (*Pongo*=solitary) + Activity (*Aotus*= nocturnal):

   'DQI + System (P=S) + Act (A=N)' tab (Table S21)

j. Model = DQI (log) + Group size (log) + Activity (*Aotus*= nocturnal):

   'DQI + Group Size + Act (A=N)' tab (Table S22)

k. Model = Diet category + System (*Pongo*=group) + Activity (*Aotus*= nocturnal):

   'Diet + System (P=G) + Act (A=N)' tab (Table S23)

l. Model = DQI (log) + System (*Pongo*=group) + Activity (*Aotus*= nocturnal):

   'DQI + System (P=G) + Act (A=N)' tab (Table S24)

3. **10k Trees consensus tree PGLS/ANOVA results (within activity pattern):**

   Tables S25-S28.xlsx

   a. Model = Diet category + System (*Pongo*=solitary) + Activity (*Aotus*=cathemeral):

      'Diet + System (P=S) + Act (A=C)' tab (Table S25)

   b. Model = Diet category + Group size (log) + Activity (*Aotus*= cathemeral):

      'Diet + Group Size + Act (A=C)' tab (Table S26)

   c. Model = DQI (log) + System (*Pongo*=solitary) + Activity (*Aotus*= cathemeral):

      'DQI + System (P=S) + Act (A=C)' tab (Table S27)

   d. Model = DQI (log) + Group size (log) + Activity (*Aotus*= cathemeral):

      'DQI + Group Size + Act (A=C)' tab (Table S28)

4. **Results across block of 1000 trees from 10kTrees (Bayesian MCMC analysis)**

   Tables S29-S32.xlsx

   a. Model = Diet category + System (*Pongo*=solitary) + Activity (*Aotus*=cathemeral):

      'Diet + System (P=S) + Act (A=C)' tab (Table S29)

   b. Model = Diet category + Group size (log) + Activity (*Aotus*= cathemeral):

‘Diet + Group Size + Act (A=C)’ tab (Table S30)

c. Model = DQI (log) + System (*Pongo*=solitary) + Activity (*Aotus*= cathemeral):

‘DQI + System (P=S) + Act (A=C)’ tab (Table S31)

d. Model = DQI (log) + Group size (log) + Activity (*Aotus*= cathemeral):

‘DQI + Group Size + Act (A=C)’ tab (Table S32)

5. Perelman and colleagues consensus tree BIC comparisons:

Tables S33-S36.xlsx

a. Model = Diet category + System (*Pongo*=solitary) + Activity (*Aotus*=cathemeral):

‘Diet + System (P=S) + Act (A=C)’ tab (Table S33)

b. Model = Diet category + Group size (log) + Activity (*Aotus*= cathemeral):

‘Diet + Group Size + Act (A=C)’ tab (Table S34)

c. Model = DQI (log) + System (*Pongo*=solitary) + Activity (*Aotus*= cathemeral):

‘DQI + System (P=S) + Act (A=C)’ tab (Table S35)

d. Model = DQI (log) + Group size (log) + Activity (*Aotus*= cathemeral):

‘DQI + Group Size + Act (A=C)’ tab (Table S36)

6. Perelman and colleagues consensus tree PGLS/ANOVA results:

Tables S37-S40.xlsx

a. Model = Diet category + System (*Pongo*=solitary) + Activity (*Aotus*=cathemeral):

‘Diet + System (P=S) + Act (A=C)’ tab (Table S37)

b. Model = Diet category + Group size (log) + Activity (*Aotus*= cathemeral):

‘Diet + Group Size + Act (A=C)’ tab (Table S38)

c. Model = DQI (log) + System (*Pongo*=solitary) + Activity (*Aotus*= cathemeral):

‘DQI + System (P=S) + Act (A=C)’ tab (Table S39)

d. Model = DQI (log) + Group size (log) + Activity (*Aotus*= cathemeral): ‘DQI + Group Size + Act (A=C)’ tab (Table S40)

Notes for Supplementary Tables

* Models constructed using maximum-likelihood value of lambda or a likelihood weighted average value of lambda across the 95% confidence interval when lambda was estimated to be 0 (see Materials and Methods for details)

† Differences in BIC (dBIC) between 2 and 6 indicate moderate evidence that the model with the lower BIC provides relatively better model fit, while values greater than 6 indicate strong evidence[105]. Clear cells: dBIC < 2; Yellow cells: 2 < dBIC < 6; Pink cells: dBIC > 6

‡ Indicates rank of model for the corresponding region according to fit (i.e. dBIC value). Only best fit models (dBIC<2) are included.

§ P-values reflect Type III analysis of variance (ANOVA) tests for each model.

‖ Maximum-likelihood estimation of lambda was 0, but this value was not used. Models were run using a likelihood weighted average value of lambda (see Materials and Methods for details)

¶ Models constructed using maximum-likelihood value of lambda (unless otherwise indicated)

** Corrected significant p value cut-off for categorical socioecological variables = 0.05/3 = 0.0167

†† Mean lambda value across the block of 1000 trees from 10kTrees (using BayesTraits MCMC analysis)

‡‡ Mean coefficient estimate across the block of 1000 trees from 10kTrees (using BayesTraits MCMC analysis)

§§ p_{MCMC} is the proportion of sampled iterations in which the explanatory parameter value exhibited the opposite sign (positive or negative) as the mean estimate

GM = grey matter; WM = white matter; NAcc = nucleus accumbens; V1 = primary visual cortex; LGN = lateral geniculate nucleus of the thalamus; MOB = main olfactory bulb; AOB = accessory olfactory bulb; Vmo = trigeminal motor nucleus of the brainstem; VII = facial nucleus of the brainstem; XII = hypoglossal nucleus of the brainstem

Appendix

Source Notes

- Stephan et al.[24,51-52]
    - *Cebus*. In 1970, data are provided for *Cebus sp.* (n=1) and *Cebus albifrons* (n=1). In 1981, data are provided for *Cebus sp.* (n=2). In 1988, the same data are provided, but the species is listed as *C. albifrons* (n=2). Here, we follow other studies[122] and assume the species is *C. albifrons*.
    - *Alouatta*. Data is provided for *Alouatta seniculus* (*n*=1) in 1970, but in 1981 there is reference to *Alouatta* sp. (*n*=2). In 1988, the same data are provided, but the species is listed as *A. seniculus* (*n*=2). Here, we follow other studies[122] and assume the species is *A. seniculus*.
    - *Callicebus moloch*. The components fail to sum to the whole in the 1981 data. This may be an error due to correction for an enlarged hydrocephalus in one specimen[122]. Here, we follow other studies[122] and utilize data for *C. moloch* from 1970.
    - *Aotus trivirgatus*. The subspecies classifications for these specimens are uncertain, given that Hershkovitz's[123] classifications were not yet published and the localities from which these specimens originated are unknown. Bauchot & Stephan[124] list alternate species names for these specimens, including *A. infulatus*, which forms a distinct clade with *A. a. azarae*[125] and is a considered by some to be conspecific with *A. a. azarae*[126].
    - For some specimens, certain regions (i.e. medulla, cerebellum, neocortex (GM+WM), LGN) were re-measured in later studies. In these cases, the more recent measurements were used (see Supplementary Data: "Brain Region Data" and "Brain Region Data Notes" tabs).
- Rilling & Insel[62-63]
    - Rilling & Insel[62-63] data appear to be based on the same specimens. In the few cases where brain volume differs between these sources, the more recent data were used in our analysis.
- Sherwood et al.[64]
    - All *Pan* and *Pongo* measurements from this study were included here. For *Gorilla*, only thalamus measurements were included since all other regions were re-measured in Barks et al.[65], but Barks et al. [65] did not include the posterior portions of the thalamus in their measurements (see below).
- Bush & Allman[54-55]
    - *Daubentonia madagascariensis* measurements were based on a T2 weighted MRI in conjunction with Nissl-stained frozen sections from the same brain.

Measurement Notes

- **Whole brain**. Stephan et al.[24,51-52] calculated total fresh brain volume (including ventricles and meninges) from fresh brain mass divided by 1.036, the density of fresh brain tissue in grams per cubic centimeter. Zilles and Rehkämper[59] follow a similar procedure, collating data from the literature to estimate species typical brain masses and correcting volumes to this species typical value. Frahm et al.[34] provide V1 measurements and their % of brain weight, so we calculated brain volumes by dividing the former by the latter and dividing by 1.036. Stimpson et al.[58] provide fresh brain weights, so we calculated brain volumes by dividing weight measurements by 1.036. Sherwood et al.[53] provide regional measurements from individuals in the Stephan and Zilles collections, so brain volumes were taken from the source materials. For a handful of individuals, brain weights or volumes were provided by the author (Sherwood 2018, personal communication). Rilling and Insel[62-63] calculate whole brain volume by separating brain tissue from surrounding tissues (cerebrospinal fluid, meninges, blood vessels, muscle, fat, and bone)[63]. Consequently, these measures may not be exactly comparable to the net volume presented by Stephan et al. Similarly, Sherwood et al.[64] measured total grey matter, white matter, and ventricular space including the cerebral hemispheres, cerebellum, midbrain, and brainstem in the measurement of brain volume. Barks et al.[65] followed a procedure similar to Sherwood et al.[64]. Barger et al.[56-57], Bauernfeind et al.[43], Bush & Allman[54-55], and de Sousa et al.[60] calculate whole brain volumes converted from previously reported or measured fresh brain weights. Additional details are not provided. Macleod et al.[61] also used these methods for the histological data set, and followed those of Rilling & Insel[62-63] for the MRI data set. Although Semendeferi and Damasio[66] present whole brain volumes, this is not comparable the other datasets, as it excludes the medulla, pons, and the 'greater part of the midbrain'. Following other studies[122], this dataset was excluded from our analysis.
- **Telencephalon**. Stephan et al.[24,51-52] include the olfactory bulbs, piriform lobe, septum, striatum, schizocortex, hippocampus, and neocortex. It is assumed that Zilles and Rehkamper[59] include the same regions.
- **Neocortex**. Stephan et al.[24,51-52] include the underlying white matter (including the corpus callosum) in their neocortex measurement. Among the transitional cortices, the insular, subgenual, cingulate, and retrosplenal regions are included. This procedure was followed by de Sousa et al.[60], who measured the neocortex and corpus callosum separately. In this case, these measurements were combined in order to be comparable to those in the Stephan et al. dataset. Rilling and Insel[63], Barks et al.[65], Zilles and Rehkamper[59], and Bush & Allman[54-55] measured neocortical grey and white matter volumes separately. Rilling and Insel[63] estimated neocortical gray matter manually tracing all grey matter in each slice before editing out subcortical grey matter, and also excluded the internal capsule from white matter measurements. Zilles and Rehkamper[59] measured the neocortex and V1 grey matter separately, so these were combined here to obtain an estimate of total neocortical grey matter. Sherwood et al.[64] provide grey matter

measurements only, and measurements included the entire cortical mantle, excluding cortex located mesial to the rhinal sulcus (including the proisocortex, cingulate gyrus, rostral insula, and temporal pole; excluding the entorhinal cortex, olfactory cortex, and hippocampus). Consequently, this region was analyzed using both: 1) grey matter measurements only (from Sherwood et al.[64]; Rilling and Insel[62]; Zilles and Rehkamper[59]; Barks et al.[65]; Bush & Allman[54-55]); and 2) combined grey and white matter measurements (from Stephan et al.[24,51-52]; de Sousa et al.[60]; Rilling and Insel[63], Zilles and Rehkamper[59]; Barks et al.[65]; Bush & Allman[54-55]). Although measurements including white matter may not be exactly equivalent since inclusion of the internal capsule varies across studies, this should represent a small proportion of overall neocortical grey and white matter volume.

- **Visual cortex**. Stephan et al.[24,51-52] include the underlying white matter. Given that the white matter borders were arbitrarily defined by the shortest distance between the borders of the grey matter in microphotographs of histological sections[34], these measurements are not included in our analyses; rather, V1 grey matter measurements in Frahm et al.[34] taken on many of the same specimens are included. Bush & Allman[54], Zilles and Rehkamper[59], and de Sousa et al.[60] also include only grey matter in their measurements. Only left hemisphere measurements are provided in de Sousa et al.[60], so these measurements were doubled. One individual was included in both Frahm et al.[34] and de Sousa et al.[60], so the latter (more recent) measurement was used. de Sousa et al.[60] identified V1 grey matter on the basis of its topological location and appearance (i.e. sub-stratification of layer 4 into multiple layers). Zilles and Rehkamper[59] and Frahm et al.[34] do not include details on delineation.
- **Insular grey matter**. Measurements for some individuals in Bauernfeind et al.[43] are from only one hemisphere – in these cases, measurements were doubled. Given that the insula is contained in the Sylvian fissure, the dorsal and inferior boundaries were defined in this study by the superior limiting sulcus (or its homologous dimple in the cortical surface) and the inferior limiting sulcus, respectively. Also, adjacency to the claustrum was an important landmark used to define insular cortex boundaries. Anteriorly, the ventral boundary was distinguished from surrounding orbitofrontal cortex based on cytoarchitecture. At the limen insulae, the ventral boundary of insular cortex was defined as the dorsal boundary of the insular branch of the piriform cortex. Posteriorly, the superior and inferior limiting sulci define the dorsal and ventral borders, and the convergence of these sulci defines its posterior extent. In Barks et al.[65], insular grey matter was defined laterally by the circular sulcus and mesially by a line linking the deepest extents of the two ends of that sulcus. Insula measurements from Semendeferi and Damasio[66] were not included here due to incomparable whole brain measures (see above).
- **Paleocortex**. Stephan et al.[24,51-52] include the retrobullar cortex (anterior olfactory nucleus), prepiriform cortex, lateral and medial olfactory tracts, internal olfactory tract,

anterior commissure, olfactory tubercle, substantia innominata, and basal nucleus of Meynert. Zilles & Rehkamper[59] define the "paleocortex" as the praepiriformis (prepiriform cortex, retrobullar cortex, olfactory tubercle) and amygdala. Accordingly, only the praepiriformis measurement from Zilles & Rehkamper[59] is comparable to the Stephan et al. measurements.

- **Hippocampus**. Stephan et al. include the pre- and postcommissural hippocampus (i.e. paraterminal and supracallosal gyri). Barger et al.[57] include the cornu ammonis, dentate gyrus, prosubiculum, subiculum proper, and presubiculum. Sherwood et al.[64] and Barks et al.[65] include the dentate gyrus, hippocampus proper, and subiculum. Details are not provided by Zilles & Rehkamper[59].
- **Schizocortex**. Stephan et al.[24,51-52] include the entorhinal, perirhinal, presubicular, and parasubicular cortices and the underlying white matter.
- **Striatum**. Stephan et al.[24,51-52] include the caudate nucleus, putamen, nucleus accumbens, and parts of the internal capsule running through the striatum. For the remaining sources (Barger et al.[57]; Sherwood et al.[64]; Zilles & Rehkamper[59]), the striatum was defined as including the caudate nucleus and putamen, but excluding the nucleus accumbens. Barks et al.[65] do not provide details, but are assumed to also only include the caudate and the putamen. Consequently, this region was analyzed using: 1) all measurements; and 2) only measurements including the nucleus accumbens from Stephan et al.[24,51-52].
- **Thalamus**. In Sherwood et al.[64], the main body of the thalamus was bordered medially by the third ventricle, dorsally by the lateral ventricles, and laterally by the internal capsule. The medial and lateral geniculate nuclei were included in outlines of the thalamus. Barks et al.[65] did not include the posterior portions of the thalamus in their measurements (Sherwood 2018, personal communication) so these measurements were not included in our analyses. Stephan et al.[24,51-52] do not provide details.
- **Lateral geniculate nucleus**. de Sousa et al.[60] include the entire structure, including all dorsal lateral geniculate nucleus laminae and interlaminar space. Only left hemisphere measurements are provided, so these measurements were doubled. Stephan et al.[24,51-52] and Bush & Allman[55] do not provide details, so here we assume the whole structure was included.
- **Subthalamus**. Stephan et al.[24,51-52] include the pallidum and subthalamic nucleus.
- **Septum**. Stephan et al.[24,51-52] include the septum pellicuIum, septum verum, diagonal band of Broca, bed nuclei of anterior commissure and stria terminalis. Zilles and Rehkamper[59] do not provide details.
- **Amygdala.** Stephan et al.[24,51-52] include the centro-medial complex (central and medial amygdaloid nuclei, anterior amygdaloid area including the nucleus of the lateral olfactory tract) and cortico-basolateral complex (lateral, cortical, and basal amygdaloid nuclei). Barger et al.[56-57] present volumetric data on multiple amygdaloid nuclei (central, lateral, basal, accessory basal) as well as whole amygdala. The latter was identified using: 1) the appearance of the basolateral nuclei at the rostral pole; 2) anterior sections: the internal

capsule laterally, the striated composition of the piriform cortex medially and rostrally, white matter dorsally and ventrally; 3) posterior sections: the putamen dorsally and laterally, the substantia innominata (presence of the basal nucleus of Meynert) dorsally; 4) caudal sections: the sulcus semiannularis medially, the lateral ventricle and hippocampus dorsally. Stimpson et al.[58] followed a procedure similar to Barger et al.[56-57]. The former included only the left hemisphere, so those measurements were doubled here. Details are not provided by Zilles & Rehkamper[59] or Barks et al.[65].

- **Cerebellum**. Stephan et al.[24,51-52] include the peduncles and basal pons in their overall measure. Sherwood et al.[64] also included deep nuclei and peduncles. Macleod et al.[61] excluded peduncles and did not include any brainstem structures. Given that pontine volumes are very small relative to overall cerebellum volume[61], the authors suggest these measurements are comparable to those in Stephan et al. Barks et al.[65] and Macleod et al.[61] provide measurements for the cerebellar hemispheres and vermis separately, so these were combined here. Zilles & Rehkamper[59] measured total cerebellum and pons separately, so only the former was included to make these measurements comparable to other studies. Rilling & Insel[62] do not provide details. Measurements from Semendeferi and Damasio[66] were not included here due to incomparable whole brain measures (see above).
- **Medulla**. Sherwood et al.[53] used identical anatomical boundaries as in the Stephan et al.[24,51-52] dataset. Medulla measures from the former were included here when such measures included all individuals in Stephan et al. Details are not provided by Zilles & Rehkamper[59].

Supplementary References

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